CalPEFT-SAM: Parameter-Efficient Adaptation of SAM 2 With Conformal False-Negative Control for Lesion Segmentation

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***Abstract*—Segmentation foundation models can be adapted to a new lesion type by training a small fraction of their weights. Before such a model outlines lesions, two further properties matter: whether its pixel probabilities can be trusted, and whether the share of lesion it misses can be bounded. We examine both with CalPEFT-SAM, which freezes the SAM 2.1 Hiera-Tiny image encoder, trains low-rank adapters and a small multi-scale decoder (0.565 million parameters, 2% of the model), and sets the decision threshold by conformal risk control so that the expected false-negative rate on new images is at most 10%. On Kvasir-SEG colonoscopy and BUSI breast ultrasound images, CalPEFT-SAM matched a fully fine-tuned ResNet-34 U-Net that trains 43 times more parameters (mean Dice over three seeds 0.912 versus 0.899, and 0.764 versus 0.746); the per-image difference was significant only on Kvasir-SEG, and only under the Wilcoxon test. Over 1000 random calibration–test re-splits, the mean false-negative rate stayed below 10% for all seven models compared, although single splits exceeded the target in up to 39% of cases. Replacing the encoder with SAM, MedSAM, MedSAM2 or DINOv2 did not change Dice significantly, and medical pretraining brought no gain. An auxiliary calibration loss did not measurably improve calibration at any of three weights. A stricter 5% target was impractical on BUSI: in nine of eleven runs, the threshold fell to zero, flagging the whole image, in 48% to 92% of re-splits. Code and data splits are publicly available.**

***Index Terms*—Calibration, conformal risk control, false-negative rate, foundation models, low-rank adaptation, medical image segmentation, segment anything model.**

I. Introduction

OUTLINING a lesion is a routine step in many imaging workflows. The contour of a colorectal polyp or a breast mass is used to measure the lesion, to follow it over time and to plan treatment, and it is increasingly drawn by a network rather than by hand. For these uses the two kinds of segmentation error do not cost the same. Extra pixels around a lesion are visible to the reader and easy to discard; lesion tissue that the model leaves out may simply go unnoticed. A network that outputs one mask at a threshold of 0.5 gives no indication of how much of the lesion that mask is likely to miss.

Segmentation foundation models have changed how such networks are built. The Segment Anything Model (SAM) [1] and its successor SAM 2 [2] were trained on more than a billion masks. Their features transfer to medical images, but zero-shot accuracy varies widely across modalities and structures, and some adaptation is usually needed [3], [4]. One route is to continue training the whole model on medical data, as MedSAM [5] and MedSAM2 [6] do. The other is parameter-efficient fine-tuning (PEFT): the pretrained weights stay frozen and a small number of new parameters are trained, for example low-rank adapters (LoRA) [7] in SAMed [8] or bottleneck adapters in the Medical SAM Adapter [9] and SAM2-UNet [10]. PEFT lets a model be adapted on a laptop GPU and stored as a few megabytes per task. Most of these studies report overlap measures such as the Dice coefficient. Whether the probabilities of the adapted model are calibrated, and what they imply for missed lesion tissue, is rarely examined.

Networks trained with Dice or cross-entropy losses tend to be over-confident [11], [12], and auxiliary losses have been proposed to correct this, among them the differentiable average calibration error of Barfoot et al. [13], [14]. Calibration describes probabilities on average, however; it does not tell the user where to cut. Conformal risk control (CRC) [15] addresses the cut directly. Given a held-out calibration set and any model that outputs pixel probabilities, CRC selects a threshold such that the expected false-negative rate (FNR) on a new image is at most a chosen level α. The only assumption is that calibration and test images are exchangeable. Angelopoulos et al. demonstrated the idea on polyp segmentation with PraNet [16].

PEFT and CRC fit together naturally: CRC does not care how the probabilities were produced, and PEFT keeps their production cheap. Several practical questions decide whether the combination is worth using, and we did not find a study that answers them together. Does a frozen encoder with a few hundred thousand trainable parameters lose anything against full fine-tuning when the endpoint is a risk-controlled region rather than a mask at 0.5? Does the choice of foundation encoder matter, and does pretraining on medical images help? Is an explicit calibration loss still needed when the threshold is set conformally? And how does the guarantee behave on finite data, that is, on a single split, at stricter levels of α, and on datasets that contain images without any lesion?

We address these questions on two public datasets with different imaging physics, Kvasir-SEG [17] (colonoscopy) and BUSI [18] (breast ultrasound, including normal images without lesions), with all models trained on the same splits and schedule on an 8 GB laptop GPU. Our contributions are the following.

1) CalPEFT-SAM, a simple pipeline that couples a

This work was supported by [funding information, or remove this sentence].

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frozen SAM 2.1 Hiera-Tiny encoder, LoRA, a 0.33 M-parameter multi-scale decoder and CRC thresholding. With 0.565 M trainable parameters it matched a fully fine-tuned ResNet-34 U-Net with 24.44 M on both datasets over three seeds.
2) A controlled comparison of seven models, five of which share the decoder, loss and training schedule and differ in the frozen encoder. No encoder differed significantly from SAM 2.1 Hiera-Tiny, and medical pretraining gave no benefit.
3) An empirical check of the CRC guarantee over 1000 random re-splits for every model, with an analysis of its weak points: violations on single splits, the effect of lesion-free images on the per-image FNR, and the collapse of the threshold at $\alpha = 0.05$ on BUSI.
4) An ablation showing that the auxiliary calibration loss had no measurable effect at weights of 0.1, 0.5 and 1.0, and that LoRA ranks of 4 to 16 performed alike while rank 2 did not.

The code, the data splits and the outputs of every run are publicly available.

II. Related Work

A. Foundation Models and PEFT for Medical Segmentation

SAM [1] couples a ViT image encoder, pretrained as a masked autoencoder, with a prompt encoder and a light mask decoder; it was trained on SA-1B, a set of 11 million images and 1.1 billion masks. SAM 2 [2] replaces the ViT with the hierarchical Hiera encoder [19], adds a memory module for video and was trained on SA-1B images together with video data that include the SA-V dataset. MedSAM [5] fine-tuned SAM ViT-B on more than 1.5 million medical image–mask pairs, and MedSAM2 [6] fine-tuned SAM 2.1 on over 455 000 3D image–mask pairs and 76 000 video frames. Both are designed to be prompted with a bounding box.

PEFT keeps the pretrained encoder fixed. SAMed [8] applies LoRA to the image encoder of SAM and fine-tunes the mask decoder for multi-organ CT segmentation. The Medical SAM Adapter [9] inserts adapter modules [20] into the encoder, and SAM2-UNet [10] places adapters in the Hiera encoder of SAM 2, replaces the mask decoder with a U-shaped decoder and reports strong results on polyp segmentation. Dutt et al. [21] compared 17 PEFT methods on six medical datasets for classification and text-to-image generation and found them most useful when training data are scarce. Our architecture is close to these designs. Our interest is different: we compare encoders under one recipe and measure calibration and risk control rather than overlap alone.

B. Calibration of Segmentation Networks

Guo et al. [11] showed that modern classifiers are over-confident and popularized the expected calibration error (ECE) [22]. In segmentation, calibration is usually measured over pixels, so the ECE is dominated by the large and easy background. Mehrtash et al. [12] found that networks trained with the Dice loss are less well calibrated than those trained with cross-entropy, and that model ensembling improves calibration. The average calibration error (ACE) gives every non-empty bin the same weight and is therefore more sensitive to the sparsely populated bins around lesion boundaries [13]. Barfoot et al. turned the ACE into a differentiable auxiliary loss, mL1-ACE, and reported lower ACE at little cost in Dice on four 3D datasets [13], [14]. Pretraining has been linked to better calibrated predictions [23], and recent non-convolutional architectures are among the best calibrated image classifiers [24]. These findings raise the question of whether a calibration loss is needed at all when most of the network is a frozen pretrained transformer.

C. Conformal Prediction for Segmentation

Conformal prediction provides distribution-free, finite-sample guarantees under exchangeability [25], [26]. Risk-controlling prediction sets [27], learn-then-test [28] and CRC [15] extend it from coverage to general bounded losses such as the FNR. The first two bound the loss with high probability over the calibration data, whereas CRC bounds its expected value. For segmentation, Angelopoulos et al. [15] controlled the per-image FNR of PraNet on pooled polyp datasets. Brunekreef et al. [29] calibrated groups of similar pixels jointly to use few calibration images efficiently; Mossina et al. [30] built prediction sets for semantic segmentation that contain the true mask at a chosen confidence level; Davenport [31] derived inner and outer confidence sets for polyp masks; Mossina and Friedrich [32] obtained prediction sets by dilating predicted masks; and Tan et al. [33] bounded the FNR of 3D lesion segmentation for several networks, SAM-Med3D among them. These studies take the segmentation model as given. We look at the other side of the pipeline: how the way the model is adapted changes the region that is needed to meet the same guarantee.

III. Method

Fig. 1 gives an overview. For an RGB image $x \in [0,1]^{3\times256\times256}$ the network outputs a map $p \in [0,1]^{256\times256}$ of foreground probabilities. It is trained on the training split only; a separate calibration split is used once, after training, to choose the threshold $t^\star$ of the reported region $\{u : p_u \geq t^\star\}$, where u indexes pixels. The name refers to its two calibration steps, a calibration term in the training loss and the conformal calibration of the threshold; only the second turned out to matter (Section V-E).

A. Frozen Encoder With Low-Rank Adaptation

The image is resized bilinearly to 512×512 pixels and normalized with the ImageNet mean and standard deviation before it enters the encoder. We use the image encoder of SAM 2.1 Hiera-Tiny (Hiera-T) [2]: a Hiera trunk [19] with four stages of 1, 2, 7 and 2 blocks (96, 192, 384 and 768 channels) followed by the feature pyramid neck of SAM 2, which returns 256-channel maps at strides 4, 8 and 16, i.e., 128×128, 64×64 and 32×32 for a 512-pixel input. The prompt encoder, mask decoder and memory modules of

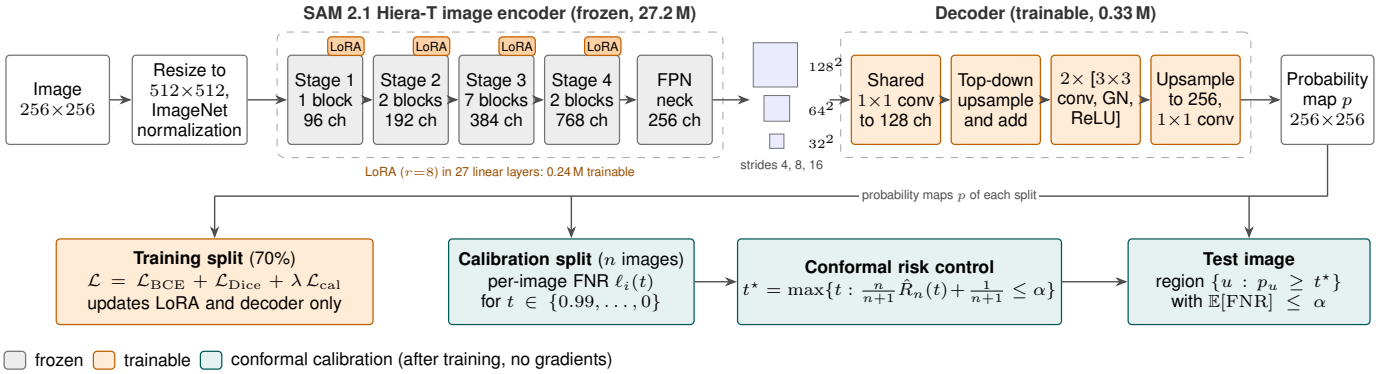


Fig. 1. Overview of CalPEFT-SAM. The SAM 2.1 Hiera-T image encoder is frozen; LoRA updates in its attention and stage-transition projections and a small top-down decoder are the only trained parts. After training, the probability maps of the calibration split are used once to choose the threshold $t^\star$ by conformal risk control, and the region $\{u : p_u \geq t^\star\}$ is reported for each test image. GN: group normalization.

SAM 2 are not used, so the model needs no prompts and runs fully automatically.

All 27.2 M encoder weights are frozen. Each attention block of the trunk contains two linear layers, the joint query–key–value projection and the output projection, and three further linear layers project the residual path where the channel width changes between stages. We wrap each of these 27 layers with a LoRA update [7],

$$h = Wx + \frac{\alpha_{\mathrm{L}}}{r} BAx, \tag{1}$$

where $W \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}$ is frozen, $A \in \mathbb{R}^{r \times d_{\mathrm{in}}}$ and $B \in \mathbb{R}^{d_{\mathrm{out}} \times r}$ are trained, $r = 8$ and $\alpha_{\mathrm{L}} = 16$. A is initialized with Kaiming-uniform values and B with zeros, so that training starts exactly from the pretrained encoder. The multilayer perceptrons of the trunk and the neck are left unchanged. The adapters add 0.237 M parameters.

B. Multi-Scale Decoder

The decoder is a small top-down feature pyramid [34]. One 1×1 convolution, shared by all levels, projects every encoder map to 128 channels. Starting from the coarsest level, the running map is upsampled bilinearly to the next finer level and added to its projection. Two 3×3 convolutions, each followed by group normalization with eight groups [35] and a ReLU, refine the resulting stride-4 map, which is then upsampled to 256×256 and passed through a 1×1 convolution to give one logit per pixel. The decoder has 0.329 M parameters. Together with the adapters, 0.565 M parameters are trained, 2.03% of the 27.78 M parameters of the model.

The same decoder is used with the plain ViT encoders that we compare in Section IV-B. There, the outputs of blocks 3, 6, 9 and 12 serve as four maps of equal resolution, and the top-down path reduces to a sum of their projections.

C. Training Objective

The network is trained with

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}} + \lambda \mathcal{L}_{\mathrm{cal}}, \tag{2}$$

where $\mathcal{L}_{\mathrm{BCE}}$ is the binary cross-entropy averaged over pixels and $\mathcal{L}_{\mathrm{Dice}} = 1 - (2\sum_u p_u y_u + \epsilon)/(\sum_u p_u + \sum_u y_u + \epsilon)$ is the soft Dice loss [36], both computed over all pixels of a mini-batch, with $y_u \in \{0, 1\}$ the label and $\epsilon = 10^{-6}$. The term $\mathcal{L}_{\mathrm{cal}}$ is a differentiable calibration error. Each probability p_u is given a soft membership in $M = 15$ equal-width bins with centers c_m,

$$w_{um} = \frac{\exp\left(-|p_u - c_m|/\tau\right)}{\sum_{k=1}^{M} \exp\left(-|p_u - c_k|/\tau\right)}, \qquad \tau = 0.05. \tag{3}$$

With the soft bin counts $S_m = \sum_u w_{um}$, the mean confidence and the observed foreground frequency of bin m are $\bar{e}_m = \sum_u w_{um} p_u / S_m$ and $\bar{o}_m = \sum_u w_{um} y_u / S_m$, and

$$\mathcal{L}_{\mathrm{cal}} = \sum_{m=1}^{M} \frac{S_m}{\sum_k S_k} \left|\bar{e}_m - \bar{o}_m\right|, \tag{4}$$

again over all pixels of the mini-batch. The term is inspired by mL1-ACE [13], [14], but it uses a softmax kernel for the soft binning and weights each bin by its soft occupancy. It is therefore a smooth surrogate of the pixel-level ECE rather than of the ACE, a distinction we return to in Section VI. We use $\lambda = 0.1$ by default and vary it in Section V-E.

D. Conformal Control of the False-Negative Rate

Let (x_i, y_i), $i = 1, \ldots, n$, be the calibration images with their masks and p_i the corresponding probability maps. For a threshold t, the per-image FNR is the fraction of lesion pixels left outside the region $\{u : p_{iu} \geq t\}$,

$$\ell_i(t) = \frac{\left|\{u :\ y_{iu} = 1,\ p_{iu} < t\}\right|}{\left|\{u :\ y_{iu} = 1\}\right|}, \tag{5}$$

with $\ell_i(t) = 0$ when the mask is empty. The loss lies in $[0, 1]$ and does not increase as t decreases. With the empirical risk $\hat{R}_n(t) = \frac{1}{n}\sum_{i=1}^{n} \ell_i(t)$ and following [15], we choose

$$t^\star = \max\left\{t \in \mathcal{T} :\ \frac{n}{n+1}\hat{R}_n(t) + \frac{1}{n+1} \leq \alpha\right\} \tag{6}$$

over a grid $\mathcal{T}$ of 200 equally spaced values from 0.99 down to 0, and set $t^\star = 0$ (the whole image) if no grid value satisfies the condition. If the calibration images and a new test image are exchangeable, then

$$\mathbb{E}\left[\ell_{n+1}(t^\star)\right] \leq \alpha, \tag{7}$$

where the expectation is taken over both the calibration set and the test image [15, Thm. 1]. We use $\alpha = 0.1$ unless stated otherwise. The bound is marginal. It concerns the average FNR over random draws of calibration and test data, not the FNR of a particular image or of a particular split. The convention $\ell_i = 0$ for empty masks also means that lesion-free images, such as the normal cases of BUSI, count as perfectly covered; Section V-C shows what this implies for the images that do contain a lesion. As the price of the guarantee we report the region size, the mean fraction of image pixels with $p_u \geq t^\star$. At the same guarantee, a smaller region means a more useful model.

IV. Experimental Setup

A. Datasets and Splits

Kvasir-SEG [17] contains 1000 colonoscopy images of polyps with masks that were drawn manually and verified by an experienced gastroenterologist; the image resolution ranges from 332×487 to 1920×1072 pixels. BUSI [18] contains 780 breast ultrasound images (437 benign, 210 malignant and 133 normal) from 600 women, with an average size of about 500×500 pixels. The masks of the normal images are empty. A few BUSI images come with more than one mask file, and we used only the first. Pawłowska et al. [37] reported that 155 BUSI images are copies of other images in the collection, that some duplicates carry different masks, and that some images show the axilla, biopsy needles or on-screen annotations. We used the dataset as distributed, as most published studies do, and return to this point in Section VI.

Each dataset was split once at random (seed 42) into 70% training, 15% calibration and 15% test images, giving 700/150/150 images for Kvasir-SEG and 546/117/117 for BUSI. The split was not stratified. The BUSI calibration set contains 69 benign, 22 malignant and 26 normal images and the test set 57, 40 and 20; this imbalance matters in Section V-C. Images and masks were resized to 256×256 pixels, intensities were scaled to $[0, 1]$ and masks were binarized at 0.5. Training images were augmented with horizontal and vertical flips (each with probability 0.5), rotation by a random multiple of 90° and, with probability 0.5, a change of brightness and contrast by factors drawn uniformly from $[0.8, 1.2]$. The calibration split was used only to choose $t^\star$. No model selection or early stopping was done on the calibration or test split.

B. Compared Models

Table I lists the seven models. Two are convolutional baselines without a foundation encoder: a U-Net [38] trained from scratch, with four resolution levels of 32 to 256 channels and batch normalization (1.93 M parameters), and a U-Net with an ImageNet-pretrained [39] ResNet-34 encoder [40] from the segmentation_models_pytorch library [41], in which all 24.44 M parameters are fine-tuned (U-Net-R34). The other five use the recipe of Section III with different frozen encoders: SAM 2.1 Hiera-T (the default, i.e., CalPEFT-SAM); MedSAM2 [6], whose released checkpoint has the same Hiera-T architecture; SAM ViT-B [1]; MedSAM [5], which has the ViT-B architecture of SAM; and DINOv2 ViT-S/14 [42], a self-supervised encoder trained on natural images without any masks. The ViT encoders, loaded through timm [43], receive 512×512 inputs (518×518 for DINOv2, whose patch size is 14), and LoRA is applied to the query–key–value and output projections of their 12 attention blocks (24 layers). For SAM ViT-B, blocks 3, 6, 9 and 12 are its global-attention blocks; for DINOv2, their outputs are taken after the final layer normalization. MedSAM inputs are min–max normalized per image, as in its original preprocessing, all other encoders use ImageNet normalization, and the U-Net trained from scratch receives the $[0, 1]$ images directly.

C. Training Details

All models were trained for 40 epochs with AdamW [44] (default β values, weight decay 0.01) and a cosine learning-rate schedule without restarts [45], using automatic mixed precision [46]. The learning rate was 10^{-3} for the U-Net trained from scratch and for the SAM family, and 10^{-4} for U-Net-R34 and DINOv2; in a preliminary run at 10^{-3}, DINOv2 reached a Dice of only 0.558 on Kvasir-SEG and 0.474 on BUSI. The batch size was 2 for the models with a foundation encoder and 8 for the two U-Nets. The weights after the last epoch were evaluated, without checkpoint selection. The training seed, which controls the initialization of new layers, the augmentation and the batch order, was 42, and CalPEFT-SAM and U-Net-R34 were also trained with seeds 43 and 44 on both datasets, with the same split. All experiments ran on one laptop with an NVIDIA GeForce RTX 5050 Laptop GPU (8 GB) using PyTorch 2.12 (nightly build, CUDA 12.8).

D. Evaluation

Dice and intersection over union (IoU) were computed for each test image at a threshold of 0.5 and averaged over images [47]. An empty prediction on an empty mask scores 1, and any predicted pixel on an empty mask scores 0, so the 20 normal test images of BUSI reward correct rejection. Calibration was measured on all test pixels pooled together (9.83 M pixels for Kvasir-SEG and 7.67 M for BUSI) with 15 equal-width bins. The ECE weights the gap between mean confidence and observed foreground frequency in each bin by the fraction of pixels in that bin [11], [22]. The ACE averages the gaps of the non-empty bins with equal weight [13], which gives the sparsely populated intermediate bins, mostly lesion boundaries, the same influence as the background. For the ablation we also report the ROI-ECE, the ECE computed only over pixels that belong to the lesion or are predicted as lesion at 0.5. For CRC we report the test FNR, the region size and $t^\star$.

A single calibration–test split yields one draw of a random quantity, so we also evaluated CRC over 1000 random re-splits. For each model, the calibration and test images (300 for Kvasir-SEG, 234 for BUSI) were pooled, randomly permuted and divided into two parts of the original sizes. The threshold was chosen on the first part, and the FNR and region size were measured on the second. The same 1000 permutations were used for every model of a dataset, and the trained network was

TABLE I
MODELS COMPARED, THEIR TRAINING SETTINGS AND THEIR COMPUTATIONAL COST

		Parameters (M)					Cost (same GPU)		
Model	Pretraining data	Input	Total	Trained	LR	Batch	Mem. (GB)	Step (s)	Infer. (ms)
U-Net, from scratch	none	256	1.93	1.928	10^{-3}	8	0.22	0.050	5.5
U-Net-R34, full fine-tuning	ImageNet-1k	256	24.44	24.436	10^{-4}	8	TBD	TBD	TBD
DINOv2 ViT-S/14 + LoRA	LVD-142M (self-sup.)	518	22.62	0.566	10^{-4}	2	0.65	0.109	37.3
SAM ViT-B + LoRA	SA-1B	512	90.51	0.837	10^{-3}	2	2.18	0.212	56.5
MedSAM ViT-B + LoRA	SA-1B, medical	512	90.51	0.837	10^{-3}	2	2.18	0.212	63.1
SAM 2.1 Hiera-T + LoRA (CalPEFT-SAM)	SA-1B, videos	512	27.78	0.565	10^{-3}	2	1.00	0.106	33.8
MedSAM2 Hiera-T + LoRA	SA-1B, videos, medical	512	27.78	0.565	10^{-3}	2	1.00	0.108	36.5

Input: encoder input size in pixels. "Medical": fine-tuned on medical data after natural-image pretraining. Mem.: peak GPU memory in training and Step: time per training step, both at batch size 2; Infer.: time per test image, one image per forward pass; all with mixed precision on an RTX 5050 Laptop GPU (8 GB), as medians of 10 repetitions after 3 warm-up iterations.

not changed. We report the mean, standard deviation and 95th percentile of the test FNR, the violation rate (the fraction of re-splits with FNR above α) and the mean region size.

Differences in Dice were tested per image with the two-sided Wilcoxon signed-rank test [48]. In the seven-model comparison, every model was compared with CalPEFT-SAM (seed 42), and the six p-values of a dataset were adjusted with the method of Holm [49]. For the three-seed comparison, the per-image Dice of each model was first averaged over the seeds. On these averages we report the Wilcoxon test, the paired t-test and a 95% bootstrap confidence interval of the mean difference with 5000 resamples of test images [50].

V. RESULTS

A. Comparison With Full Fine-Tuning

In Table II, CalPEFT-SAM had the higher mean Dice on both datasets and was ahead in each of the three seeds (Dice differences of $+0.0130$, $+0.0108$ and $+0.0141$ on Kvasir-SEG and $+0.0076$, $+0.0340$ and $+0.0114$ on BUSI), while training 0.565 M instead of 24.44 M parameters.

Per image, the evidence is weaker. On Kvasir-SEG, CalPEFT-SAM was better on 92 of the 150 test images and worse on 58. The median difference was small ($+0.0024$), but most differences were positive, and the Wilcoxon test gave $p = 0.0057$, whereas the paired t-test ($p = 0.089$) and the bootstrap interval of the mean ($[-0.0016, +0.0271]$) did not exclude zero. The mean is moved by a few images on which one of the two models fails: five images differ by more than 0.3 in Dice, in both directions. On BUSI, the two models tied on 20 images: 14 normal images on which both returned an empty mask with every seed, two further normal images, and four lesions that both missed with every seed. CalPEFT-SAM was better on 45 images and worse on 52. The median difference was zero and none of the tests was significant. We therefore read Table II as parity with full fine-tuning at 43 times fewer trained parameters, with a small and seed-consistent advantage on Kvasir-SEG, rather than as a clear win.

Both models had a low pixel ECE. The ACE was lower for CalPEFT-SAM on both datasets, although with a large seed-to-seed spread on Kvasir-SEG (± 0.021). Both models met the FNR target on average, and CalPEFT-SAM needed a slightly smaller region on Kvasir-SEG and the same region on BUSI.

B. Choice of the Frozen Encoder

Table III lists all seven models trained with seed 42. Training from scratch on 700 or 546 images clearly did not suffice: the U-Net reached a Dice of only 0.643 on Kvasir-SEG and 0.573 on BUSI, far below every pretrained model (Holm-adjusted $p < 10^{-12}$ against CalPEFT-SAM), and needed about twice the region or more to meet the FNR target. Among the five frozen encoders, Dice ranged from 0.904 to 0.912 on Kvasir-SEG and from 0.733 to 0.786 on BUSI. None of the differences to CalPEFT-SAM was significant after Holm correction; the smallest adjusted p-value was 0.40 on Kvasir-SEG and 1.0 on BUSI.

Medical pretraining did not help. MedSAM2 was 0.003 better than its natural-image counterpart on Kvasir-SEG and 0.027 worse on BUSI. MedSAM was 0.002 better than SAM ViT-B on Kvasir-SEG and 0.028 worse on BUSI (Wilcoxon $p = 0.35$ and 0.66, uncorrected). None of these differences was significant, but on BUSI both point in the direction opposite to what one would expect if medical fine-tuning transferred to breast ultrasound. DINOv2, which was never trained with a segmentation mask, obtained the highest Dice and the smallest CRC region on BUSI, but also the highest ECE on Kvasir-SEG among the pretrained models.

All pretrained models were reasonably calibrated, with ECE values of 0.0176–0.0213 on Kvasir-SEG and 0.0223–0.0279 on BUSI, against 0.0461 for the U-Net trained from scratch on Kvasir-SEG. CalPEFT-SAM had the lowest ECE of all seven models on both datasets and the lowest ACE on BUSI. Calibration and region size were not aligned, however. On BUSI, CalPEFT-SAM needed the largest region of the five frozen encoders (11.73% against 9.18%–10.57%), and DINOv2 the smallest. The region at a fixed FNR depends on how well the probabilities rank lesion pixels above background, which is a different property from calibration.

C. Validity of the FNR Guarantee

Fig. 2 shows the distribution of the test FNR over the 1000 re-splits. For every model on both datasets the mean was below the target, 0.0918–0.0942 on Kvasir-SEG and 0.0778–0.0937 on BUSI, and the same held for the extra seeds and all ablation runs (the largest of all 28 means was 0.0942). Nor was it overly conservative: for losses without jumps, the expected FNR of

TABLE II
CALPEFT-SAM VERSUS THE FULLY FINE-TUNED U-NET-R34 OVER THREE TRAINING SEEDS

Dataset	Model	Trained (M)	Dice ↑	IoU ↑	ECE ↓	ACE ↓	CRC FNR	CRC region (%) ↓
Kvasir-SEG	U-Net-R34	24.44	0.8989 ± 0.0061	0.8380 ± 0.0078	0.0185 ± 0.0026	0.1183 ± 0.0079	0.0939 ± 0.0003	15.40 ± 0.05
	CalPEFT-SAM	0.565	$\mathbf{0.9115} \pm 0.0045$	$\mathbf{0.8565} \pm 0.0057$	$\mathbf{0.0156} \pm 0.0024$	$\mathbf{0.0926} \pm 0.0214$	0.0937 ± 0.0002	$\mathbf{14.88} \pm 0.11$
BUSI	U-Net-R34	24.44	0.7464 ± 0.0110	0.6809 ± 0.0078	0.0243 ± 0.0031	0.1689 ± 0.0109	0.0903 ± 0.0025	$\mathbf{11.23} \pm 0.50$
	CalPEFT-SAM	0.565	$\mathbf{0.7641} \pm 0.0065$	$\mathbf{0.6947} \pm 0.0066$	$\mathbf{0.0226} \pm 0.0011$	$\mathbf{0.1299} \pm 0.0134$	0.0908 ± 0.0015	11.26 ± 0.56

	Per-image Dice difference, CalPEFT-SAM minus U-Net-R34 (each image averaged over the three seeds)					
Dataset	Mean	Median	Better / worse / tied	Wilcoxon p	Paired t-test p	95% bootstrap CI
Kvasir-SEG ($n = 150$)	$+0.0126$	$+0.0024$	92 / 58 / 0	0.0057	0.089	$[-0.0016,\ +0.0271]$
BUSI ($n = 117$)	$+0.0177$	0.0000	45 / 52 / 20	0.80	0.10	$[-0.0031,\ +0.0389]$

Top: mean $\pm$ standard deviation over seeds 42, 43 and 44. CRC values are means over 1000 calibration–test re-splits at $\alpha = 0.1$ (Section IV-D); the region is the percentage of image pixels in $\{p \geq t^\star\}$. Bold marks the better mean of the two models; CRC FNR is not marked because both models meet the target.

TABLE III
ALL SEVEN MODELS WITH TRAINING SEED 42

	Segmentation and calibration (threshold 0.5)						CRC at $\alpha = 0.1$, 1000 re-splits		
Model	Dice ↑	IoU ↑	ECE ↓	ACE ↓	ΔDice	p_{Holm}	Mean FNR	FNR $> \alpha$ (%)	Region (%) ↓
Kvasir-SEG									
U-Net, from scratch	0.6432	0.5190	0.0461	0.1188	-0.2656	4.0×10^{-24}	0.0918	33.1	30.20
U-Net-R34, full fine-tuning	0.8958	0.8347	0.0183	0.1128	-0.0130	0.14	0.0942	33.9	15.44
DINOv2 ViT-S/14 + LoRA	0.9044	**0.8555**	0.0213	0.1260	-0.0045	0.40	0.0935	36.2	**14.84**
SAM ViT-B + LoRA	0.9063	0.8489	0.0181	**0.0874**	-0.0025	0.40	0.0937	33.6	14.98
MedSAM ViT-B + LoRA	0.9085	0.8527	0.0181	0.0901	-0.0003	0.95	0.0937	34.6	14.85
SAM 2.1 Hiera-T + LoRA (CalPEFT-SAM)	0.9088	0.8547	**0.0176**	0.0929	–	–	0.0936	33.9	14.97
MedSAM2 Hiera-T + LoRA	**0.9118**	0.8552	0.0180	0.1125	$+0.0030$	0.95	0.0937	34.2	15.00
BUSI									
U-Net, from scratch	0.5729	0.4876	0.0260	0.1291	-0.1926	1.0×10^{-13}	0.0778	12.5	34.84
U-Net-R34, full fine-tuning	0.7578	0.6891	0.0237	0.1678	-0.0076	1.00	0.0924	35.9	10.89
DINOv2 ViT-S/14 + LoRA	**0.7863**	**0.7133**	0.0228	0.1417	$+0.0209$	1.00	0.0937	39.0	**9.18**
SAM ViT-B + LoRA	0.7612	0.6909	0.0250	0.1501	-0.0042	1.00	0.0919	37.9	10.21
MedSAM ViT-B + LoRA	0.7334	0.6671	0.0279	0.1542	-0.0320	1.00	0.0913	34.8	10.57
SAM 2.1 Hiera-T + LoRA (CalPEFT-SAM)	0.7654	0.6982	**0.0223**	**0.1156**	–	–	0.0893	33.1	11.73
MedSAM2 Hiera-T + LoRA	0.7380	0.6689	0.0256	0.1770	-0.0274	1.00	0.0928	36.5	10.07

ΔDice: mean per-image Dice difference to CalPEFT-SAM. p_{Holm}: two-sided Wilcoxon signed-rank test against CalPEFT-SAM, Holm-adjusted over the six comparisons of a dataset. FNR $> \alpha$: percentage of re-splits in which the test FNR exceeded α. Best Dice, IoU, ECE, ACE and region per dataset in bold.

CRC is at least $\alpha - 2/(n+1)$ [15], i.e., 0.087 for Kvasir-SEG and 0.083 for BUSI, and only the U-Net trained from scratch on BUSI fell below this value (0.078).

Single splits were another matter. The standard deviation of the test FNR across re-splits was 0.015–0.019 on Kvasir-SEG and 0.026–0.034 on BUSI, the 95th percentiles reached 0.120–0.126 and 0.137–0.148, and the target was exceeded in 33.1%–36.2% of the Kvasir-SEG re-splits and in 12.5%–39.0% of the BUSI re-splits (Table III). This is consistent with a bound on the expectation, but it matters whenever CRC is reported on a single split. The fixed split of BUSI was an unfavorable draw: on it, the test FNR at $t^\star$ was above 0.1 for six of the seven models, up to 0.163 for MedSAM (0.113 for CalPEFT-SAM and 0.142 for U-Net-R34). Its calibration part contains more normal images than its test part (26 against 20), which contribute zero loss, and fewer malignant cases (22 against 40). Malignant lesions were missed more often; at $t^\star$, CalPEFT-SAM had an FNR of 0.151 on the malignant and 0.126 on the benign test images. The re-splits show that the procedure itself is not at fault.

A second caveat concerns the empty masks of BUSI. Because lesion-free images count as zero loss, the guarantee over all images is looser for the images that matter. Averaged over the re-splits at $\alpha = 0.1$, the FNR on images with a lesion was 0.111 for CalPEFT-SAM and 0.115 for U-Net-R34 (0.111–0.117 for all six pretrained models), against 0.089 and 0.092 over all images. Since every lesion-free image has $\ell = 0$, $\mathbb{E}[\ell] = \pi\, \mathbb{E}[\ell \mid \text{lesion}]$, where π is the fraction of images that contain a lesion. A bound of α for lesion images is therefore obtained by targeting

$$\alpha' = \pi\, \alpha, \tag{8}$$

which is 0.083 for BUSI ($\pi = 647/780$). Kvasir-SEG contains no lesion-free images, so the question does not arise there.

D. Risk Level and Region Size

Table IV shows how the region grows as α is tightened. On Kvasir-SEG the cost is gradual for both models: lowering α from 0.20 to 0.05 enlarged the region of CalPEFT-SAM from 12.51% to 17.26% of the image and that of U-Net-R34 from 13.14% to 17.82%. On BUSI, $\alpha = 0.05$ was impractical. The threshold of CalPEFT-SAM fell to zero, i.e., to the whole image, in 91.5% of the re-splits, and the same happened in

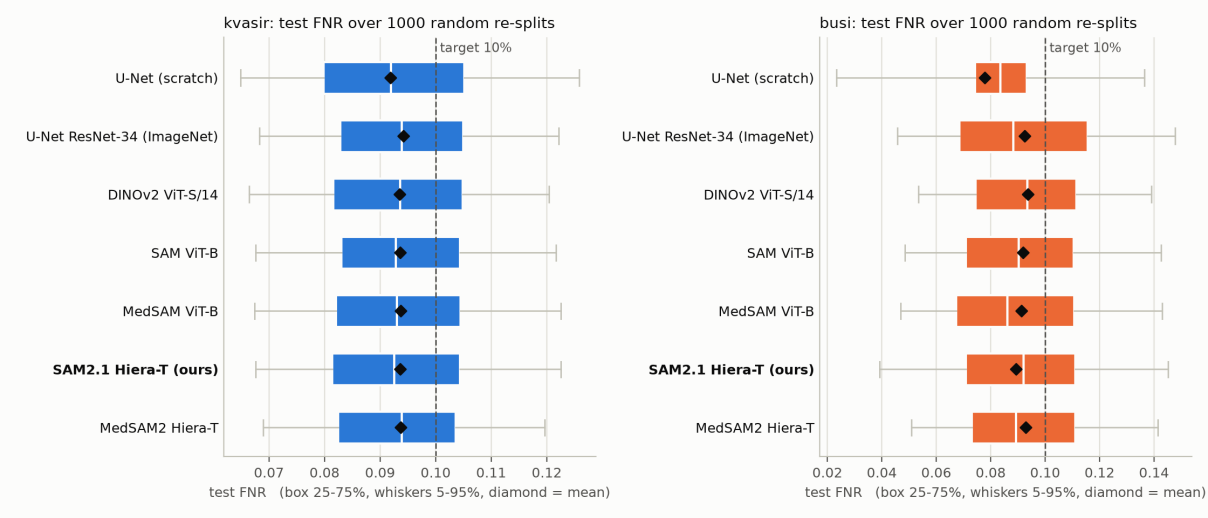
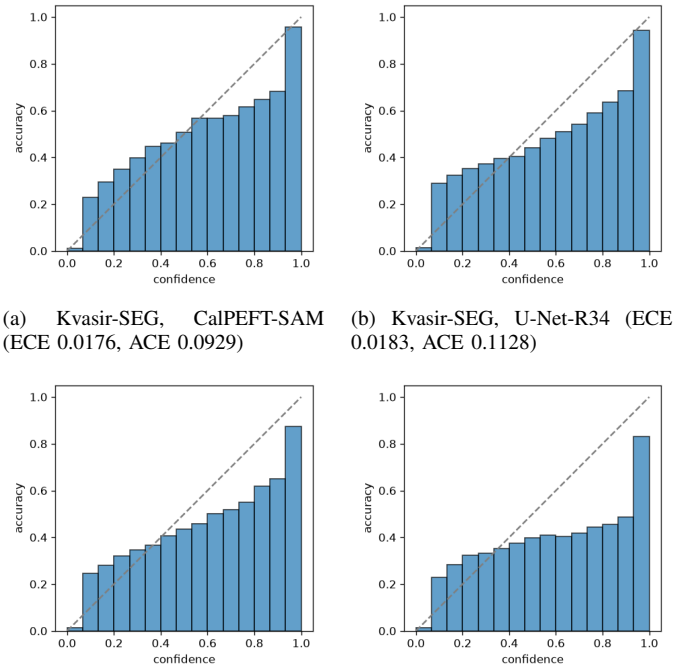


Fig. 2. Test FNR of CRC at $\alpha = 0.1$ over 1000 random calibration–test re-splits for the seven models (seed 42). Boxes span the 25th to 75th percentile, whiskers the 5th to 95th percentile, and diamonds mark the mean. The dashed line is the target. "SAM2.1 Hiera-T (ours)" is CalPEFT-SAM.

TABLE IV
MEAN TEST FNR AND REGION SIZE OVER 1000 RE-SPLITS FOR DIFFERENT RISK LEVELS α (SEED 42)

	Kvasir-SEG		BUSI	
α	CalPEFT-SAM	U-Net-R34	CalPEFT-SAM	U-Net-R34
0.05	0.0441 / 17.26	0.0434 / 17.82	0.0066 / 92.82	0.0315 / 28.24
0.10	0.0936 / 14.97	0.0942 / 15.44	0.0893 / 11.73	0.0924 / 10.89
0.15	0.1423 / 13.62	0.1432 / 14.17	0.1441 / 8.01	0.1443 / 7.85
0.20	0.1872 / 12.51	0.1905 / 13.14	0.1941 / 6.48	0.1943 / 6.70

Each cell: mean test FNR / mean region size in percent of the image.



(a) Kvasir-SEG, CalPEFT-SAM (ECE 0.0176, ACE 0.0929)

(b) Kvasir-SEG, U-Net-R34 (ECE 0.0183, ACE 0.1128)

(c) BUSI, CalPEFT-SAM (ECE 0.0223, ACE 0.1156)

(d) BUSI, U-Net-R34 (ECE 0.0237, ACE 0.1678)

Fig. 3. Pixel-level reliability diagrams on the test sets (seed 42, 15 bins). Bars show the observed foreground frequency in each confidence bin; the dashed line marks perfect calibration.

47.9% to 92.3% of the re-splits for nine of the eleven BUSI runs (mean regions of 55.5% to 93.3%). The exceptions were the U-Net trained from scratch, which still needed 57.7% of the image (with $t^\star = 0$ in 16.4% of the re-splits), and U-Net-R34 with seed 42 (28.2%, never zero); with seeds 43 and 44, U-Net-R34 fell to zero in 62.9% and 56.3% of the re-splits.

The cause is the risk left at the smallest positive threshold. At $\alpha = 0.05$, (6) requires an empirical risk below 0.042 on 117 calibration images, but even with every pixel above $p = 0.005$ included, the mean per-image FNR over the pooled BUSI images was 0.050–0.057 for the three CalPEFT-SAM runs and 0.044–0.045 for U-Net-R34 with seeds 43 and 44 (0.013 with seed 42). In these five runs, the ten worst-covered of the 188 lesions carried 56%–69% of this residual. Whether a strict target can be met thus depends on a few hard lesions and can change between seeds of the same model, which Dice does not reveal (the three U-Net-R34 runs had Dice values of 0.736–0.758).

E. Calibration and the Auxiliary Loss

Fig. 3 shows reliability diagrams for CalPEFT-SAM and U-Net-R34. Both models put nearly all pixels into the first and the last bin. The first, background bin is well calibrated and keeps the pixel ECE small; the last bin is close to the diagonal on Kvasir-SEG but over-confident on BUSI (observed frequency of about 0.88 for CalPEFT-SAM and 0.83 for U-Net-R34). In the intermediate bins, which hold mostly boundary pixels, the observed frequency rises much more slowly than the confidence. Pixels predicted at about 0.1 were lesion roughly a quarter of the time, and pixels predicted at 0.8 only about 60% of the time for CalPEFT-SAM on Kvasir-SEG, so the models are over-confident on both sides of 0.5 in the boundary zone. On BUSI, the observed frequency of U-Net-R34 stayed below 0.5 up to a confidence of 0.9, which

TABLE V
ABLATION OF THE CALIBRATION WEIGHT λ AND THE LORA RANK r FOR CALPEFT-SAM ON KVASIR-SEG

λ	r	Trained (M)	Dice	ECE	ACE	ROI-ECE	Region (%)	p
0	8	0.565	0.9121	0.0148	0.0864	0.1073	14.82	0.38
0.1	8	0.565	0.9088	0.0176	0.0929	0.1226	14.97	–
0.5	8	0.565	0.9127	0.0158	0.0841	0.1116	14.78	0.90
1.0	8	0.565	0.9005	0.0172	0.0994	0.1200	15.14	0.089
0.1	2	0.388	0.8933	0.0194	0.1312	0.1323	15.05	0.002
0.1	4	0.447	0.9159	0.0136	0.0832	0.1011	14.95	0.77
0.1	16	0.802	0.9092	0.0168	0.1125	0.1169	14.85	0.52
Default, seed 43			0.9167	0.0129	0.0711	–	14.76	0.32
Default, seed 44			0.9091	0.0164	0.1139	–	14.91	0.31

Seed 42 unless stated; the default is $\lambda = 0.1$, $r = 8$. Region: CRC region at $\alpha = 0.1$, mean over 1000 re-splits; the mean test FNR was 0.0936–0.0939 in every row. p: Wilcoxon signed-rank test of per-image Dice against the default. The two seed rows show the variation caused by the training seed alone.

explains its high ACE.

The upper part of Table V varies the weight λ of the calibration term. Without the term ($\lambda = 0$), the ECE (0.0148), ACE (0.0864) and ROI-ECE (0.1073) were all lower than with the default $\lambda = 0.1$ (0.0176, 0.0929 and 0.1226). With $\lambda = 0.5$, the ECE and ROI-ECE lay between the two and the ACE was slightly lower (0.0841), and $\lambda = 1.0$ gave a lower Dice (0.9005) and a higher ACE (0.0994). None of the Dice differences to the default was significant ($p = 0.38$, 0.90 and 0.089). The ECE and ACE differences are no larger than those caused by the training seed alone: across seeds 42, 43 and 44 of the default model, the ECE ranged from 0.0129 to 0.0176 and the ACE from 0.071 to 0.114, which covers every λ setting. In the form we used, the calibration term thus had no measurable effect on calibration, on Dice or on the CRC region, which stayed between 14.78% and 15.14% of the image.

F. LoRA Rank

The lower part of Table V varies the LoRA rank. Ranks 4 and 16 performed like rank 8 (Dice 0.9159 and 0.9092 against 0.9088; $p = 0.77$ and 0.52), and both lie within the seed-to-seed range of the default (0.9088–0.9167). Rank 4 even gave the best Dice, ECE, ACE and ROI-ECE of the ablation, with 0.447 M trained parameters. Rank 2 was clearly worse, with a Dice of 0.8933 ($p = 0.002$) and an ACE of 0.131. Because the decoder alone has 0.329 M parameters, the number of trained parameters only doubles from rank 2 to rank 16 (0.388 M to 0.802 M), and we kept $r = 8$ as a safe default.

G. Computational Cost

In Table I, CalPEFT-SAM peaked at 1.00 GB of GPU memory during training and needed 0.106 s per training step and 33.8 ms per test image. A 40-epoch run on Kvasir-SEG took about 30 min for CalPEFT-SAM and about 12 min for U-Net-R34, which uses a larger batch. The ViT-B encoders took twice as long per training step and 1.7 to 1.9 times as long per test image, and they needed 2.18 GB. The trained weights of CalPEFT-SAM occupy 2.3 MB in 32-bit precision, against 98 MB for U-Net-R34, which matters when one model per task or per site has to be stored and distributed. The price of the frozen encoder is paid at inference: the U-Net trained from scratch needed 5.5 ms per image, six times less than CalPEFT-SAM.

H. Qualitative Results

Fig. 4 shows test examples. In the first Kvasir-SEG example, U-Net-R34 assigned high probability to a large area above and to the right of the polyp, and this area entered its region; CalPEFT-SAM added only two small blobs. In the second, the polyp reaches the image border, and the region of CalPEFT-SAM covers the lower-left part of the lesion that U-Net-R34 left out. On BUSI, the low thresholds make the regions extend beyond the lesion margin, as in the last row, and both models placed a small spurious region on the normal image in the third row.

VI. DISCUSSION

For these two tasks, most of the machinery of model adaptation could be dropped without measurable loss. The encoder stayed frozen, 98% of the weights received no gradient and no calibration loss was needed. What remained, namely LoRA, a small decoder and a conformal threshold, matched a fully fine-tuned CNN in overlap, in calibration and in the region needed for a 10% FNR guarantee, and it trained in about half an hour on a laptop GPU. Its advantage over full fine-tuning was small and consistent across seeds, but it did not hold under every statistical test, and we do not claim more than parity.

The choice of encoder mattered less than we expected. Five encoders that differ in architecture, in size (22 M to 90 M parameters) and in pretraining (self-supervised natural images, promptable segmentation of natural images, and medical fine-tuning) gave Dice values within 0.008 of each other on Kvasir-SEG and within 0.053 on BUSI, with no significant difference from CalPEFT-SAM. The medically fine-tuned encoders were not significantly better than the natural-image models they started from, and on BUSI they were numerically worse. Medical fine-tuning may have shaped the encoders around box-prompted segmentation with their original mask decoders, both of which our setting replaces, or BUSI may simply be too small and noisy to show a benefit; our data cannot separate the two. SAM 2.1 Hiera-T remains a reasonable default because it was the best calibrated encoder and among the cheapest, not because it segments better.

Why did the calibration loss not help? The adapted models were already well calibrated where most pixels are, in the background bin. A plausible explanation is that the term we used weights bins by occupancy, so that its value and its gradient are dominated by this bin. The remaining miscalibration lies in the sparsely populated boundary bins (Fig. 3), which an occupancy-weighted term hardly sees. The uniformly weighted mL1-ACE of Barfoot et al. [14] targets exactly these bins and may behave differently; we did not test it. More basically, CRC needs probabilities that rank lesion pixels above background, not calibrated ones, and the two properties were decoupled in our results (Section V-B).

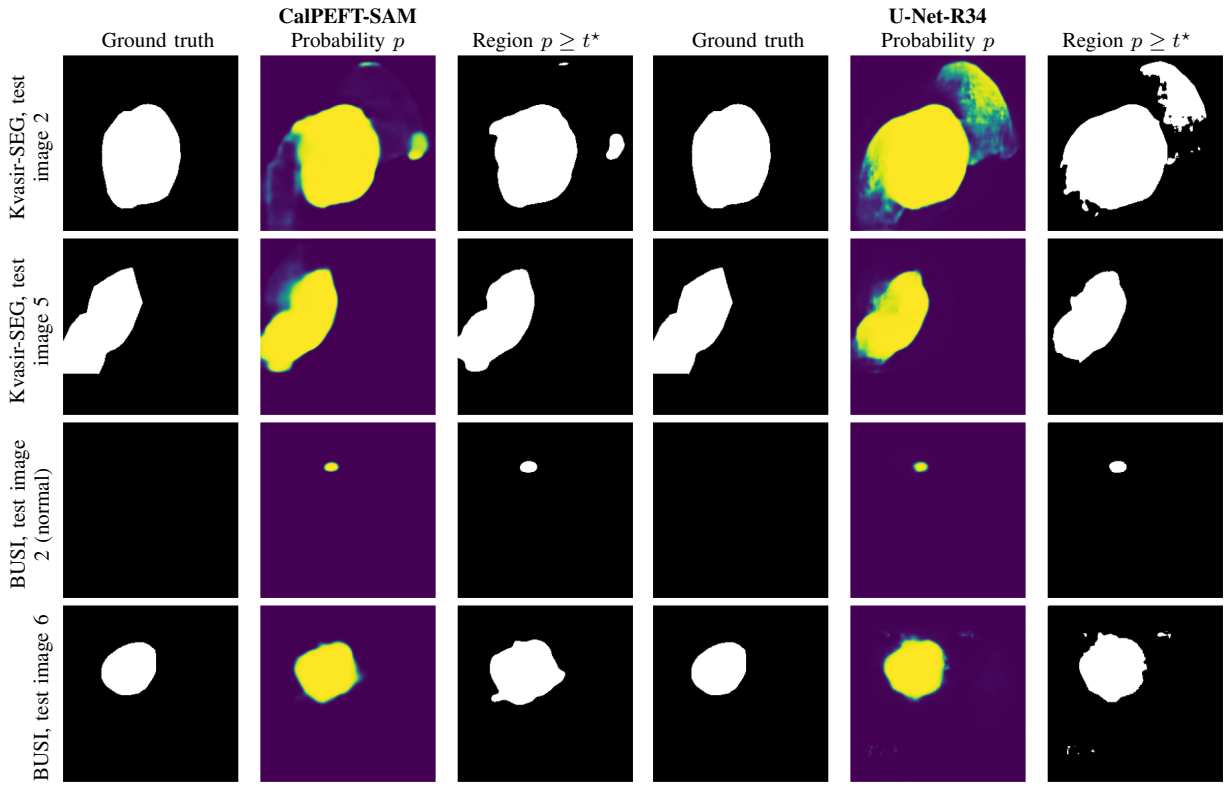


Fig. 4. Test examples (seed 42), chosen from the first six test images of each dataset, for which the outputs of every run are saved. Each strip shows the ground-truth mask, the predicted probability map and the CRC region $\{p \geq t^\star\}$ at $\alpha = 0.1$, with $t^\star = 0.67$ (CalPEFT-SAM) and $t^\star = 0.48$ (U-Net-R34) on Kvasir-SEG and $t^\star = 0.02$ and $t^\star = 0.06$ on BUSI.

The guarantee should be worded carefully in clinical reports. Besides being marginal (Section III-D), it says nothing about how often a lesion is missed entirely. Where a statement about the realized split is needed, risk-controlling prediction sets or learn-then-test [27], [28] give high-probability bounds at the cost of a larger region. Datasets with lesion-free images need the adjustment (8), or calibration on lesion images only, if the guarantee is meant for patients who have a lesion. Finally, the collapse at $\alpha = 0.05$ on BUSI shows that a threshold cannot recover lesion tissue to which the model gives almost no probability; strict targets need models that miss fewer lesions, or an image-level safeguard in addition to the pixel-level threshold.

Our study has several limitations. First, we used random, non-stratified splits of public datasets. BUSI contains duplicate images, some with different masks [37], and Kvasir-SEG provides no patient identifiers, so duplicates or images of the same patient may fall into both the training and the test set. This would inflate absolute scores, probably for all models alike, but we cannot rule out an effect on the comparisons. Second, only CalPEFT-SAM and U-Net-R34 were trained with three seeds. The encoder comparison and the ablation use one seed, and the seed-to-seed spread of the default model (Dice 0.909–0.917 on Kvasir-SEG) is as large as many of the differences in Tables III and V. Third, hyperparameters were set per model family and not tuned, apart from the learning rate of DINOv2, and U-Net-R34 was trained with a larger batch and hence fewer optimizer steps. Fourth, the evaluation is two-dimensional, at 256×256 pixels, on two single-source datasets without an external test set, and our absolute Dice values are not directly comparable with results reported on other splits or resolutions. Under a shift between calibration and deployment data, exchangeability fails and the CRC guarantee no longer holds. Fifth, the calibration term that we tested weights bins by occupancy, and our negative result does not automatically extend to the uniformly weighted mL1-ACE.

VII. Conclusion

A frozen SAM 2.1 Hiera-T encoder with LoRA and a small decoder, 0.565 M trained parameters in total, segmented polyps and breast lesions as well as a fully fine-tuned ResNet-34 U-Net, with a lower ACE on both datasets. Combined with conformal risk control, it yielded regions whose mean FNR stayed below the 10% target over 1000 re-splits. Neither medical pretraining of the encoder nor an auxiliary calibration loss improved the results. Two findings deserve attention in future work: single calibration splits can violate the target, especially on small datasets with lesion-free images, and a 5% target was impractical on BUSI for nearly every model and seed.

Data and Code Availability

Kvasir-SEG and BUSI are publicly available from their original sources [17], [18]. The code, the fixed data splits, the per-run metrics, the per-image results and the CRC curves needed to reproduce all tables and figures are available at https://github.com/MinhazulHasan/CalPEFT-SAM-Vision.

ACKNOWLEDGMENT

[Acknowledgments, or remove this section.]

REFERENCES

[1] A. Kirillov *et al.*, "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4015–4026.

[2] N. Ravi *et al.*, "SAM 2: Segment anything in images and videos," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[3] M. A. Mazurowski, H. Dong, H. Gu, J. Yang, N. Konz, and Y. Zhang, "Segment anything model for medical image analysis: An experimental study," *Med. Image Anal.*, vol. 89, 2023, art. no. 102918.

[4] Y. Huang *et al.*, "Segment anything model for medical images?" *Med. Image Anal.*, vol. 92, 2024, art. no. 103061.

[5] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, "Segment anything in medical images," *Nature Commun.*, vol. 15, 2024, art. no. 654.

[6] J. Ma *et al.*, "MedSAM2: Segment anything in 3D medical images and videos," arXiv:2504.03600, 2025.

[7] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[8] K. Zhang and D. Liu, "Customized segment anything model for medical image segmentation," arXiv:2304.13785, 2023.

[9] J. Wu *et al.*, "Medical SAM adapter: Adapting segment anything model for medical image segmentation," *Med. Image Anal.*, vol. 102, 2025, art. no. 103547.

[10] X. Xiong *et al.*, "SAM2-UNet: Segment anything 2 makes strong encoder for natural and medical image segmentation," *Vis. Intell.*, vol. 4, 2026, art. no. 2.

[11] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 70, 2017, pp. 1321–1330.

[12] A. Mehrtash, W. M. Wells, C. M. Tempany, P. Abolmaesumi, and T. Kapur, "Confidence calibration and predictive uncertainty estimation for deep medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 12, pp. 3868–3878, 2020.

[13] T. Barfoot, L. C. Garcia-Peraza-Herrera, B. Glocker, and T. Vercauteren, "Average calibration error: A differentiable loss for improved reliability in image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, ser. Lecture Notes in Computer Science, vol. 15009, 2024, pp. 139–149.

[14] T. Barfoot, L. C. Garcia-Peraza-Herrera, S. Akcay, B. Glocker, and T. Vercauteren, "Average calibration losses for reliable uncertainty in medical image segmentation," *IEEE Trans. Med. Imag.*, 2026, early access, doi: 10.1109/TMI.2026.3673118.

[15] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[16] D.-P. Fan *et al.*, "PraNet: Parallel reverse attention network for polyp segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, ser. Lecture Notes in Computer Science, vol. 12266, 2020, pp. 263–273.

[17] D. Jha *et al.*, "Kvasir-SEG: A segmented polyp dataset," in *Proc. Int. Conf. MultiMedia Model. (MMM)*, ser. Lecture Notes in Computer Science, vol. 11962, 2020, pp. 451–462.

[18] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data Brief*, vol. 28, 2020, art. no. 104863.

[19] C. Ryali *et al.*, "Hiera: A hierarchical vision transformer without the bells-and-whistles," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 202, 2023, pp. 29 441–29 454.

[20] N. Houlsby *et al.*, "Parameter-efficient transfer learning for NLP," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 97, 2019, pp. 2790–2799.

[21] R. Dutt, L. Ericsson, P. Sanchez, S. A. Tsaftaris, and T. Hospedales, "Parameter-efficient fine-tuning for medical image analysis: The missed opportunity," in *Proc. Int. Conf. Med. Imag. Deep Learn. (MIDL)*, ser. Proc. Mach. Learn. Res., vol. 250, 2024, pp. 406–425.

[22] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proc. AAAI Conf. Artif. Intell.*, 2015, pp. 2901–2907.

[23] D. Hendrycks, K. Lee, and M. Mazeika, "Using pre-training can improve model robustness and uncertainty," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 97, 2019, pp. 2712–2721.

[24] M. Minderer *et al.*, "Revisiting the calibration of modern neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 15 682–15 694.

[25] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

[26] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023.

[27] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan, "Distribution-free, risk-controlling prediction sets," *J. ACM*, vol. 68, no. 6, pp. 43:1–43:34, 2021.

[28] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *Ann. Appl. Stat.*, vol. 19, no. 2, pp. 1641–1662, 2025.

[29] J. Brunekreef, E. Marcus, R. Sheombarsing, J.-J. Sonke, and J. Teuwen, "Kandinsky conformal prediction: Efficient calibration of image segmentation algorithms," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 4135–4143.

[30] L. Mossina, J. Dalmau, and L. Andéol, "Conformal semantic image segmentation: Post-hoc quantification of predictive uncertainty," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2024, pp. 3574–3584.

[31] S. Davenport, "Conformal confidence sets for biomedical image segmentation," arXiv:2410.03406, 2024.

[32] L. Mossina and C. Friedrich, "Conformal prediction for image segmentation using morphological prediction sets," in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, ser. Lecture Notes in Computer Science, vol. 15963, 2025, pp. 78–88.

[33] B. Tan *et al.*, "Conformal lesion segmentation for 3D medical images," arXiv:2510.17897, 2025.

[34] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 2117–2125.

[35] Y. Wu and K. He, "Group normalization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[36] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. Int. Conf. 3D Vis. (3DV)*, 2016, pp. 565–571.

[37] A. Pawłowska, P. Karwat, and N. Żołek, "Letter to the editor. Re: 'Dataset of breast ultrasound images by W. Al-Dhabyani, M. Gomaa, H. Khaled & A. Fahmy, Data in Brief, 2020, 28, 104863'," *Data Brief*, vol. 48, 2023, art. no. 109247.

[38] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, ser. Lecture Notes in Computer Science, vol. 9351, 2015, pp. 234–241.

[39] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2009, pp. 248–255.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[41] P. Iakubovskii, "Segmentation models pytorch," GitHub repository. [Online]. Available: https://github.com/qubvel/segmentation_models.pytorch, 2019.

[42] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.

[43] R. Wightman, "PyTorch image models," GitHub repository. [Online]. Available: https://github.com/huggingface/pytorch-image-models, 2019.

[44] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[45] ——, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017.

[46] P. Micikevicius *et al.*, "Mixed precision training," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[47] L. Maier-Hein *et al.*, "Metrics reloaded: Recommendations for image analysis validation," *Nature Methods*, vol. 21, no. 2, pp. 195–212, 2024.

[48] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, 1945.

[49] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[50] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.